

Federated Learning

Federated Learning is a decentralized machine learning approach that trains algorithms across multiple local devices or servers without exchanging raw data. Instead of bringing data to the model, the model goes to the data, allowing AI to learn collaboratively while preserving privacy and complying with strict data regulations.

Instead of gathering all raw data into one central cloud server to train an AI model, federated learning pushes the model out to individual edge devices (like smartphones), where it learns from the local data in those devices and only share the learned or abstract mathematical updates back to the cloud.

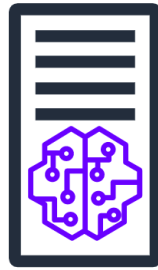
Federated Learning Vs Machine Learning

Federated Learning

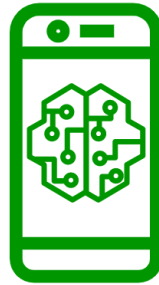
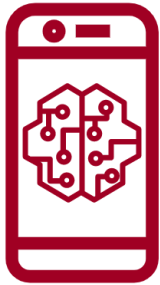
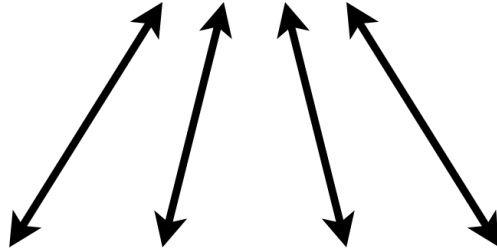
- **Data flow:** Raw data never leaves the local device. The data stays on your phone or local server, and only encrypted model updates (gradients) are sent back to the central server.
- **Model Training:** A generic "global" model is pushed to your device. It trains locally on your specific data, and the central server aggregates all users' updates to improve the global model.
- **Privacy and Security:** Highly privacy-preserving. Sensitive information like personal messages, financial transactions, or medical records is shielded from the centralized exposure.
- **Ideal for** privacy-sensitive applications like healthcare research, predictive text on mobile keyboards, financial fraud detection, and smart home IoT devices.

Machine Learning

- **Data flow:** All raw data is aggregated from various sources (phones, hospital, IoT) and moved to a centralized server or cloud data center.
- **Model Training:** The AI model is trained directly on these consolidated datasets in one location.
- **Privacy and Security:** Carries high privacy risks. Because all raw data sits in one place, it is more vulnerable to breaches, leaks, or regulatory issues (e.g., GDPR or HIPAA).
- **Use cases:** Ideal for situations where data is owned by a single entity, privacy is not a primary concern, or massive compute clusters are required to process the data.



Server coordinating
the training of a
global AI model



Devices with
local AI models